

DatabaseHelper.java

```
package com.kushmiprabin123;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {
    private static final String DATABASE_NAME = "students.db";
    private static final int DATABASE_VERSION = 1;
    private static final String TABLE_NAME = "tbl_students";
    private static final String ID = "id";
    private static final String NAME = "name";
    private static final String EMAIL = "email";
    private static final String MOBILE = "mobile";
    private static final String STATUS = "status";

    private static final String CREATE_TABLE = "CREATE TABLE " + TABLE_NAME + "(" + ID + " INTEGER
PRIMARY KEY AUTOINCREMENT, "
        + NAME + " TEXT, "
        + EMAIL + " TEXT, "
        + MOBILE + " TEXT, "
        + STATUS + " INTEGER)";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL(CREATE_TABLE);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    // Insert a Student Record

    public boolean insertStudent(String name, String email, String mobile, int status) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
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        values.put(NAME, name);
        values.put(EMAIL, email);
        values.put(MOBILE, mobile);
        values.put(STATUS, status);
        long result = db.insert(TABLE_NAME, null, values);
        db.close();
        return result>0;
    }

    // Read a Student Record
    public Cursor readStudent() {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
    }

    // Update a Student Record
    public boolean updateStudent(int id, String name, String email, String mobile, int status){
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put(NAME, name);
        values.put(EMAIL, email);
        values.put(MOBILE, mobile);
        values.put(STATUS, status);
        int result = db.update(TABLE_NAME, values, ID + "=?", new String[]{String.valueOf(id)});
        db.close();
        return result>0;
    }

    //Delete a Student Record

    public boolean deleteStudent(int id){
        SQLiteDatabase db = this.getWritableDatabase();
        int result = db.delete(TABLE_NAME, ID + "=?", new String[]{String.valueOf(id)});
        db.close();
        return result>0;
    }
}

```

MainActivity.java

```

package com.kushmiprabin123;

import static android.app.ProgressDialog.show;

import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

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import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    private Toolbar toolbar;

    DatabaseHelper dbHelper;
    EditText etId, etName, etEmail, etMobile, etStatus;
    TextView tvResult;
    Button btnCreate, btnRead, btnUpdate, btnDelete;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        toolbar=findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);

        dbHelper = new DatabaseHelper(this);
        etId = findViewById(R.id.etId);
        etName = findViewById(R.id.etName);
        etEmail = findViewById(R.id.etEmail);
        etMobile = findViewById(R.id.etMobile);
        etStatus = findViewById(R.id.etStatus);
        tvResult = findViewById(R.id.tv_result);

        btnCreate = findViewById(R.id.btn_create);
        btnRead = findViewById(R.id.btn_read);
        btnUpdate = findViewById(R.id.btn_update);
        btnDelete = findViewById(R.id.btn_delete);

        //create Student
        btnCreate.setOnClickListener(v -> {
            String name = etName.getText().toString();
            String email = etEmail.getText().toString();
            String mobile = etMobile.getText().toString();
            int status = Integer.parseInt(etStatus.getText().toString());
            boolean result = dbHelper.insertStudent(name, email, mobile,
status);

            if(result) {
                Toast.makeText(MainActivity.this, "Student Created",
Toast.LENGTH_SHORT).show();
            }else {
                Toast.makeText(MainActivity.this, "Error Creating Student",
Toast.LENGTH_SHORT).show();
            }
        })
    }
}

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    });

    //Read Student
    btnRead.setOnClickListener(v -> {
        Cursor cursor = dbHelper.readStudent();
        StringBuilder builder = new StringBuilder();
        if (cursor.getCount() == 0) {
            Toast.makeText(MainActivity.this, "No Data Found",
Toast.LENGTH_SHORT).show();
        } else {
            while (cursor.moveToNext()) {
                builder.append("ID:").append(cursor.getInt(0))
                    .append("\nNAME:").append(cursor.getString(1))
                    .append("\nEMAIL:").append(cursor.getString(2))
                    .append("\nMOBILE:").append(cursor.getString(3))
                    .append("\nSTATUS:").append(cursor.getInt(4))
                    .append("\n\n");
            }
            tvResult.setText(builder.toString());
            cursor.close();
        }
    });

    //Update Student
    btnUpdate.setOnClickListener(v -> {
        int id = Integer.parseInt(etId.getText().toString());
        String name = etName.getText().toString();
        String email = etEmail.getText().toString();
        String mobile = etMobile.getText().toString();
        int status = Integer.parseInt(etStatus.getText().toString());
        boolean result = dbHelper.updateStudent(id, name, email, mobile,
status);

        if(result){
            Toast.makeText(MainActivity.this, "Student Update",
Toast.LENGTH_SHORT).show();
        }else{
            Toast.makeText(MainActivity.this, "Update Failed",
Toast.LENGTH_SHORT).show();
        }
    });

    //Delete Student
    btnDelete.setOnClickListener(v -> {
        int id = Integer.parseInt(etId.getText().toString());
        boolean result = dbHelper.deleteStudent(id);

        if(result){
            Toast.makeText(MainActivity.this, "Student Deleted",
Toast.LENGTH_SHORT).show();
        }else{
            Toast.makeText(MainActivity.this, "Delete Failed",
Toast.LENGTH_SHORT).show();
        }
    });
}
}

```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:orientation="vertical">

    <androidx.appcompat.widget.Toolbar
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/toolbar"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/crud_operation"
        android:textSize="30sp"
        android:textAllCaps="true"
        android:layout_gravity="top"
        android:gravity="center"/>

    <EditText
        android:id="@+id/etId"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/enter_id_for_update_delete"
        android:textSize="20sp"
        android:inputType="text"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/name"
        android:textSize="20sp"/>

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/enter_your_name"
        android:textSize="20sp"
        android:inputType="text"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/email"
        android:textSize="20sp"/>

    <EditText
        android:id="@+id/etEmail"
```

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        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/enter_your_email"
        android:textSize="20sp"
        android:inputType="textEmailAddress"/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/mobile"
    android:textSize="20sp"/>

<EditText
    android:id="@+id/etMobile"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="@string/enter_your_mobile_number"
    android:textSize="20sp"
    android:inputType="phone"/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/status"
    android:textSize="20sp"/>

<EditText
    android:id="@+id/etStatus"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="@string/status_0_1"
    android:textSize="20sp"
    android:inputType="numberSigned"/>

<Button
    android:id="@+id/btn_create"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/create"/>

<Button
    android:id="@+id/btn_read"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/read"/>

<Button
    android:id="@+id/btn_update"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/update"/>

<Button
    android:id="@+id/btn_delete"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/delete"/>
```

```
<ScrollView
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/tv_result"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:paddingTop="20dp"
        android:hint="@string/result"
    />
</ScrollView>

</LinearLayout>
```